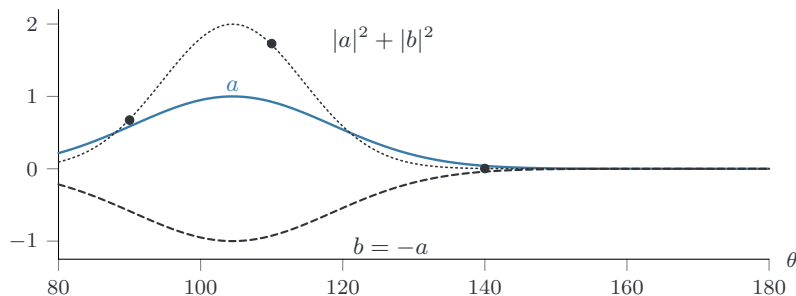
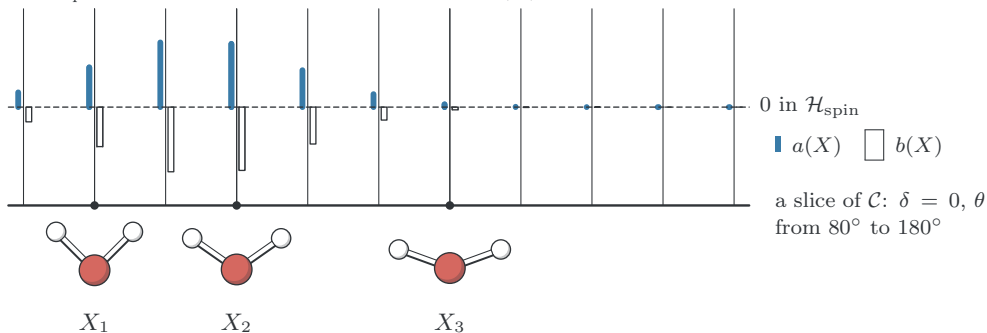


the same $\mathcal{H}_{\text{spin}}$ over every X : on each fibre the value $\Psi(X)$ as its two components a, b



$$\Psi(X) = a(X)\xi + b(X)\eta$$

$$\|\Psi\|^2 = \int_{\mathcal{C}} (|a|^2 + |b|^2) dX$$

here $b = -a$:
 $\Psi = a(\xi - \eta)$ is the proton singlet, as exchange requires of a rotation-invariant state on this slice

$\xi = \uparrow\downarrow \omega, \eta = \downarrow\uparrow \omega$ orthonormal in $\mathcal{H}_{\text{spin}} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^1$; $a(\theta)$ is an illustrative profile
the dots are $\|\Psi(X_k)\|^2$